Higher National Diploma in Engineering (Civil)
Higher National Diploma in Quantity Surveying
1st year, Second Semester Examination – 2018
ME 1205 / QS 12062 - Engineering Graphics and AutoCAD

Instructions for Candidates:	No. of questions : 04
Answer all questions	No. of pages : 03
You can use both sides of the answer sheet	Time : **03 hours**

Q1)

(i). Using **Intersection of arcs method,** draw an ellipse with major axis of 110 mm and minor axis of 80 mm.
Draw a tangent, at any point 'X' on the curve. (16 marks)

(ii). A point 'Q' on a hyperbola, is located at distances 35 mm and 25 mm away from the X & Y axes respectively. Construct the Hyperbola. (12 marks)

[Total 28 Marks]

Q2)

(i). A circle of 20 mm radius, is rolling inside of a circle of 72 mm radius without slip. Initially a point 'P' is at the contact point of two circles. Draw the locus (path) of point 'P', for one revolution of the rolling circle. (14 marks)

(ii). Draw the locus of free end of a string, when the string is tightly held and unwound one complete turn around a circle of diameter 40 mm. (String length is equal to the circumferential length of the circle). (14 marks)

[Total 28 Marks]

Q3)

(i). A vertical cylinder with diameter of 50 mm and axial height of 90 mm is completely penetrated by a horizontally positioned square prism with sides of 25 mm and axial length of 70 mm. Both axes intersect & bisect each other. All faces of prism are equally inclined to horizontal plane. Draw the projections (front view, end view and plan) showing the curves of intersections.

(17 marks)

Figure 3.1 shows the perspective (three dimentional) view of the objects while Figure 3.2 shows incomplete front view and plan.

Figure 3.1

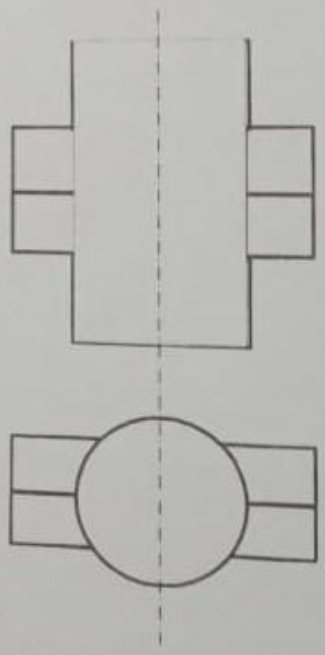


Figure 3.2

Drawing 2

Drawing 3

Q3)

(ii). A regular hexagonal pyramid with each side of base 30 mm and axial height of 60 mm rests on its base which is kept on the horizontal plane. The cutting plane Y-Y bisects the axis of pyramid and makes 40°angle with horizontal plane, as shown in Figure 3.3. Draw the development of the lateral surface of truncated pyramid when the top part is removed by cutting through plane Y-Y. (17 marks)

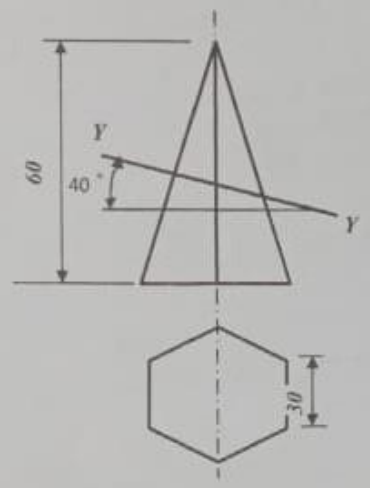
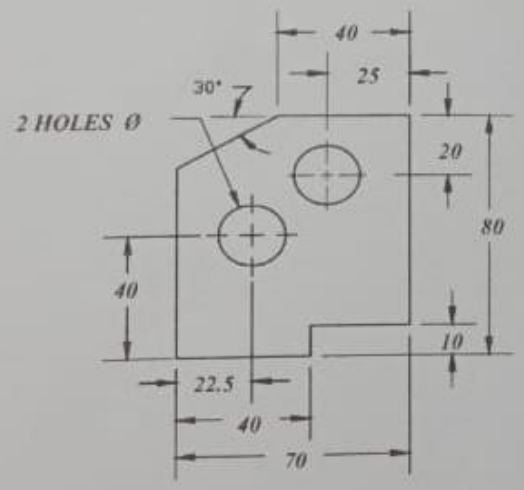


Figure 3.3

[Total 34 Marks]

Q4) Write down necessary AutoCAD commands in order to construct the object shown in the Figure 4. All dimensions are given in mm. (10 marks)



[Total 10 Marks]

Figure 4